Chapter 2

E-business strategy

Learning Objectives

In this chapter, you will learn about:

1. Introduction to the strategy of E-business
2. Web server and E-mail technologies
3. Revenue models of E-business
4. Online shopping
5. Virtual communities
6. E-marketing strategy and planning

1. Introduction to the strategy of E-business

Management issues

- How does E-business strategy differ from traditional business strategy?
- How should we integrate E-business strategy with existing business and IS strategy?
- How should we evaluate our investment priorities and returns from E-business?

Michael Porter on the Internet

- *The key question is not whether to deploy Internet technology – companies have no choice if they want to stay competitive – but how to deploy it*

Porter, M. (2001) Strategy and the Internet, *Harvard Business Review,* March 2001, 62–78.

Definitions of strategy

- What is strategy?
- A plan of action designed to achieve a particular goal.
  - ‘Defines how we will meet our objectives’
  - ‘Sets allocation of resources to meet goals’
  - ‘Selects preferred strategic option to compete within a market’
  - ‘Provides a long-term plan for the development of the organization’

Different forms of organizational strategy

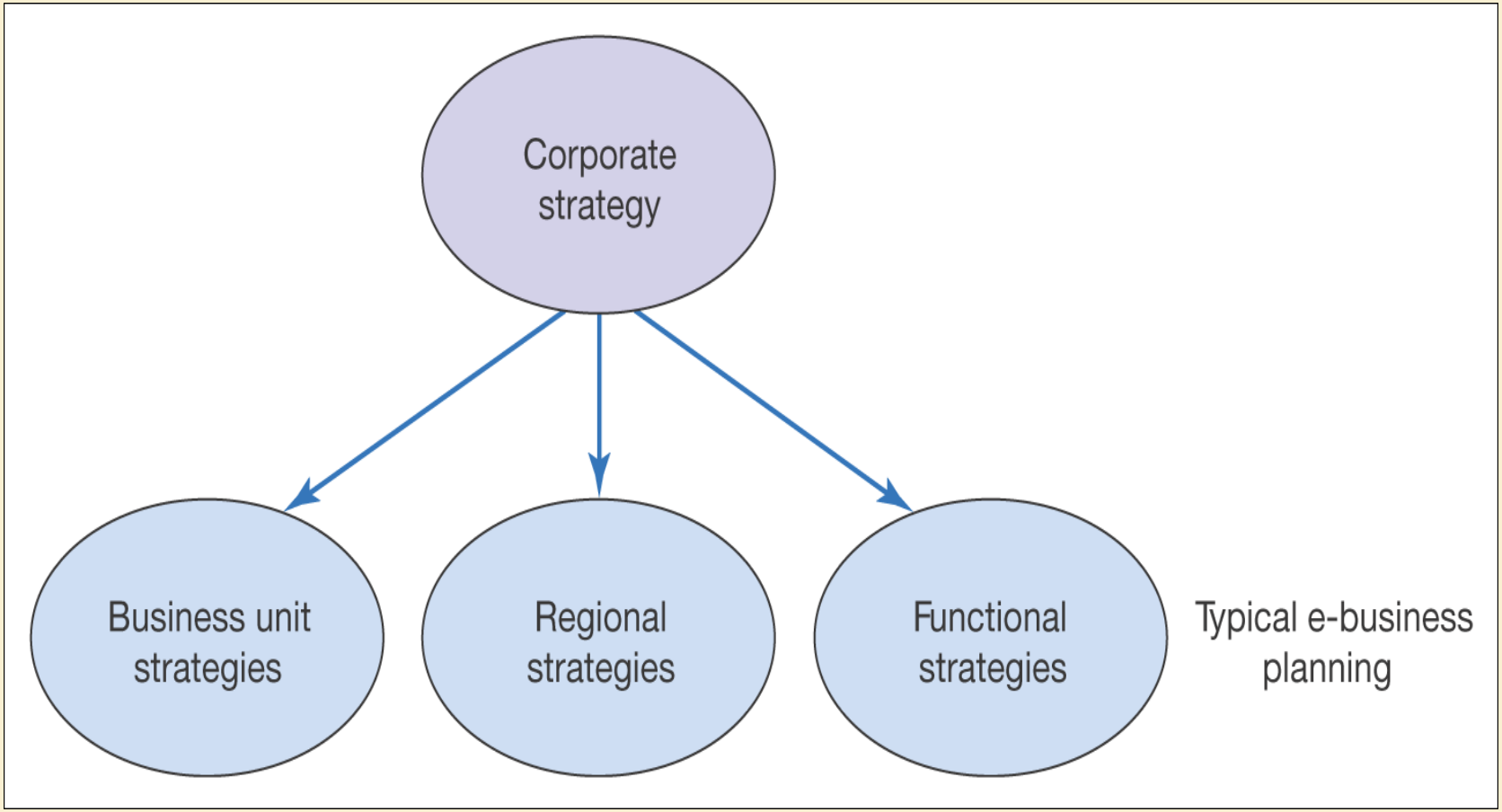


Figure 5.1 Different forms of organizational strategy

Relationship between e-business strategy and other strategies



Figure 5.2 Relationship between e-business strategy and other strategies

Sell-side e-commerce strategy

- Sell-side e-commerce is a *channel strategy*
- Objectives for online contribution percentage should push our strategy
- Channel strategy expands on *differentials*
- BUT, need to manage *channel integration*

Sell-side e-commerce strategy (Cont.)

- E-commerce strategy defines how we should
  - Hit our channel leads and sales targets
    - Acquisition, Retention, Conversion, Service, Profitability
  - Communicate the benefits of using channel
  - Prioritize products available through channel
  - Prioritize audiences targeted through channel
  - Select partners for this channel

Buy-side e-commerce strategy or e-supply chain management strategy

- Buy-side e-commerce strategy is about maximizing operational efficiencies while improving customer service quality
- Operational efficiency KPIs should drive our strategy
- Involves selection of appropriate strategic partners

Buy-side e-commerce strategy or e-supply chain management strategy (Cont.)

- Buy-side e-commerce strategy defines how we should
  - Automate internal processes
  - Link internal resource management systems with external purchasing systems
  - Prioritize suppliers / partners collaborating using this channel
  - Prioritize applications for eSCM – create a roadmap

What happens where there is no e-business strategy?

- Missed opportunities for additional sales on the sell-side and for more efficient purchasing on the buy-side
- Fall-behind competitors in delivering online services – may become difficult to catch-up, for example, Tesco, Dell
- Poor customer experience from poorly integrated channels

A generic strategy process model

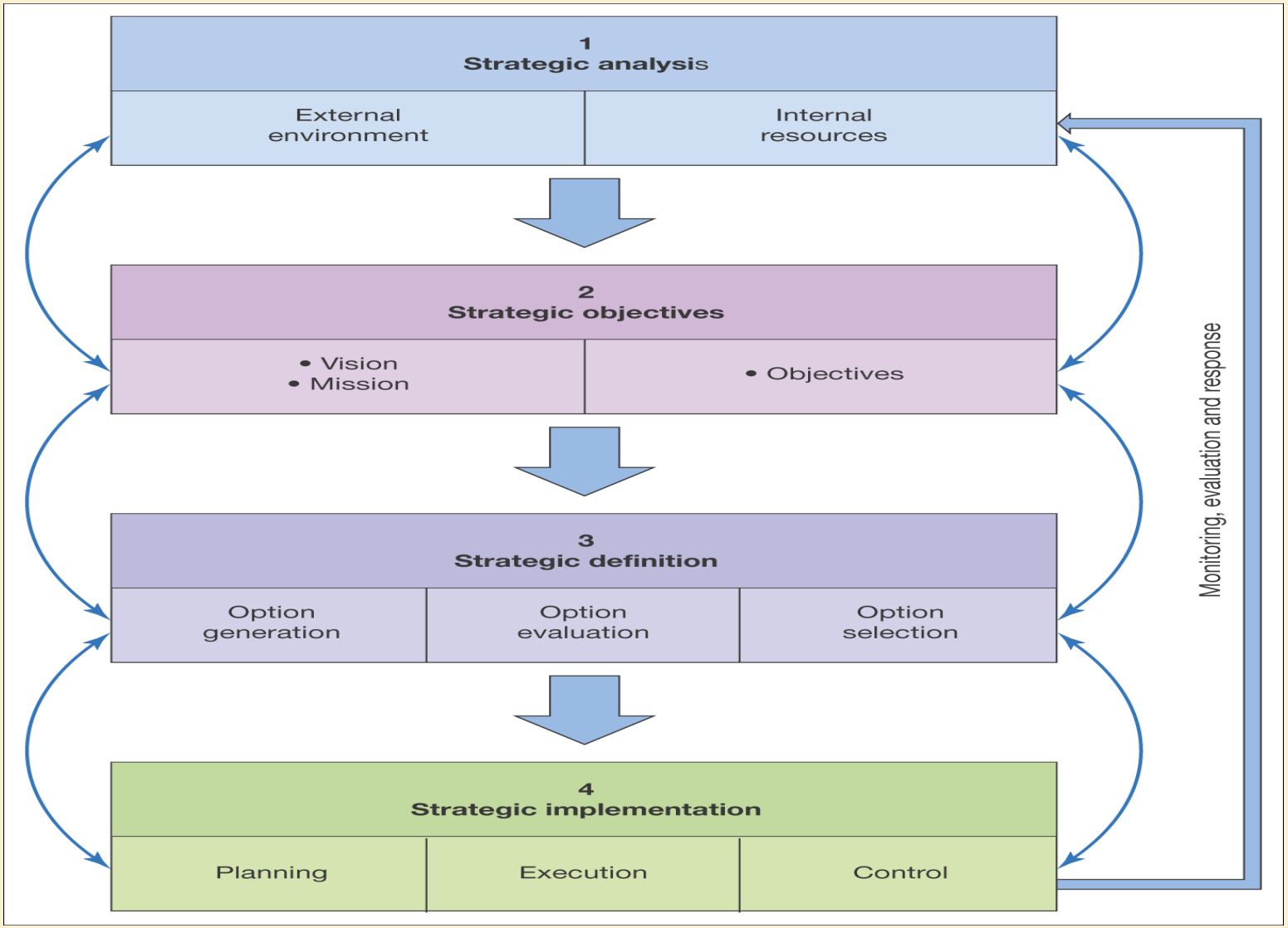


Figure 1: A generic strategy process model.

Dynamic e-business strategy model

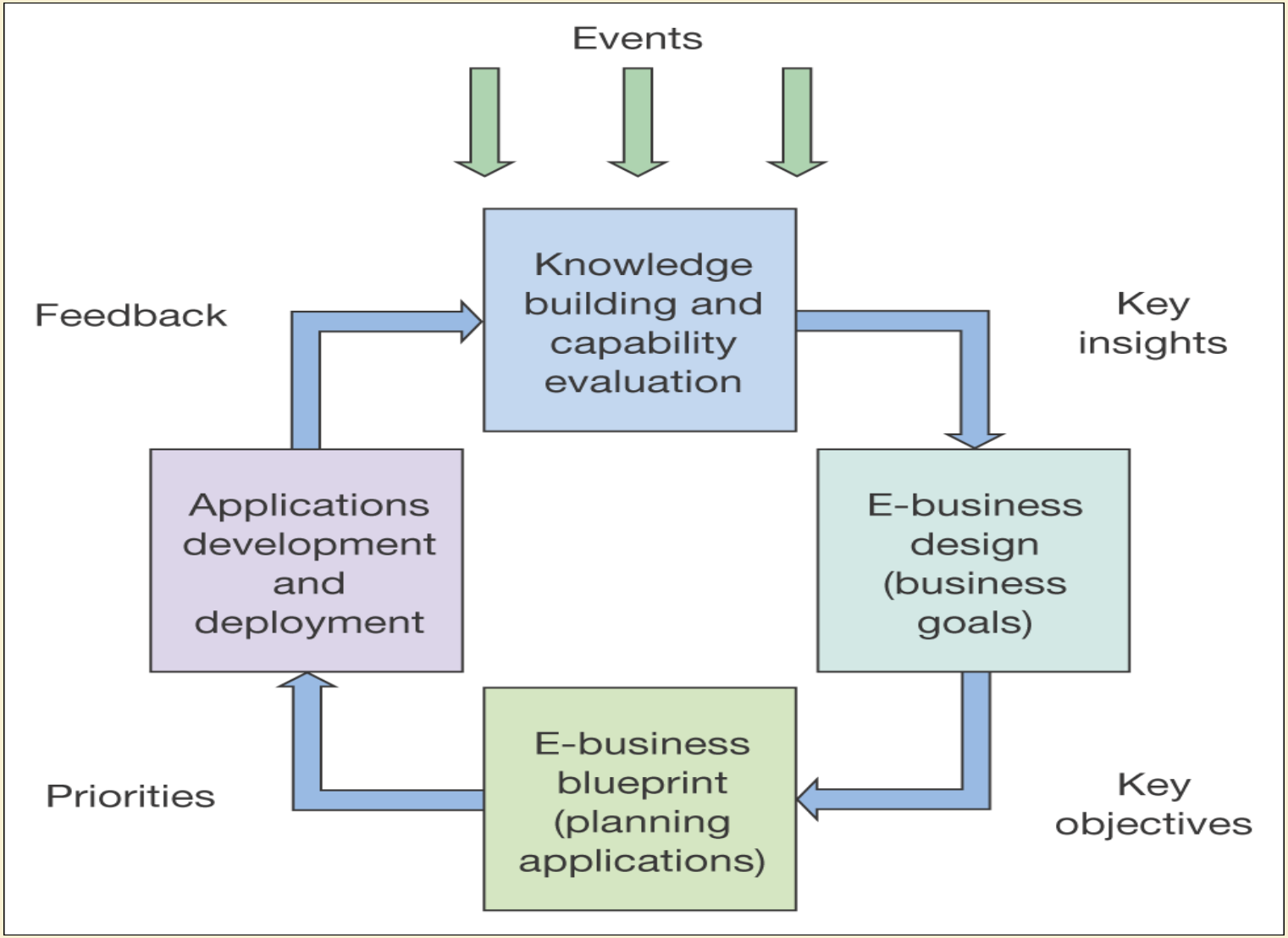


Figure 2: Dynamic e-business strategy model.

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Elements of strategic situation analysis for the e-business

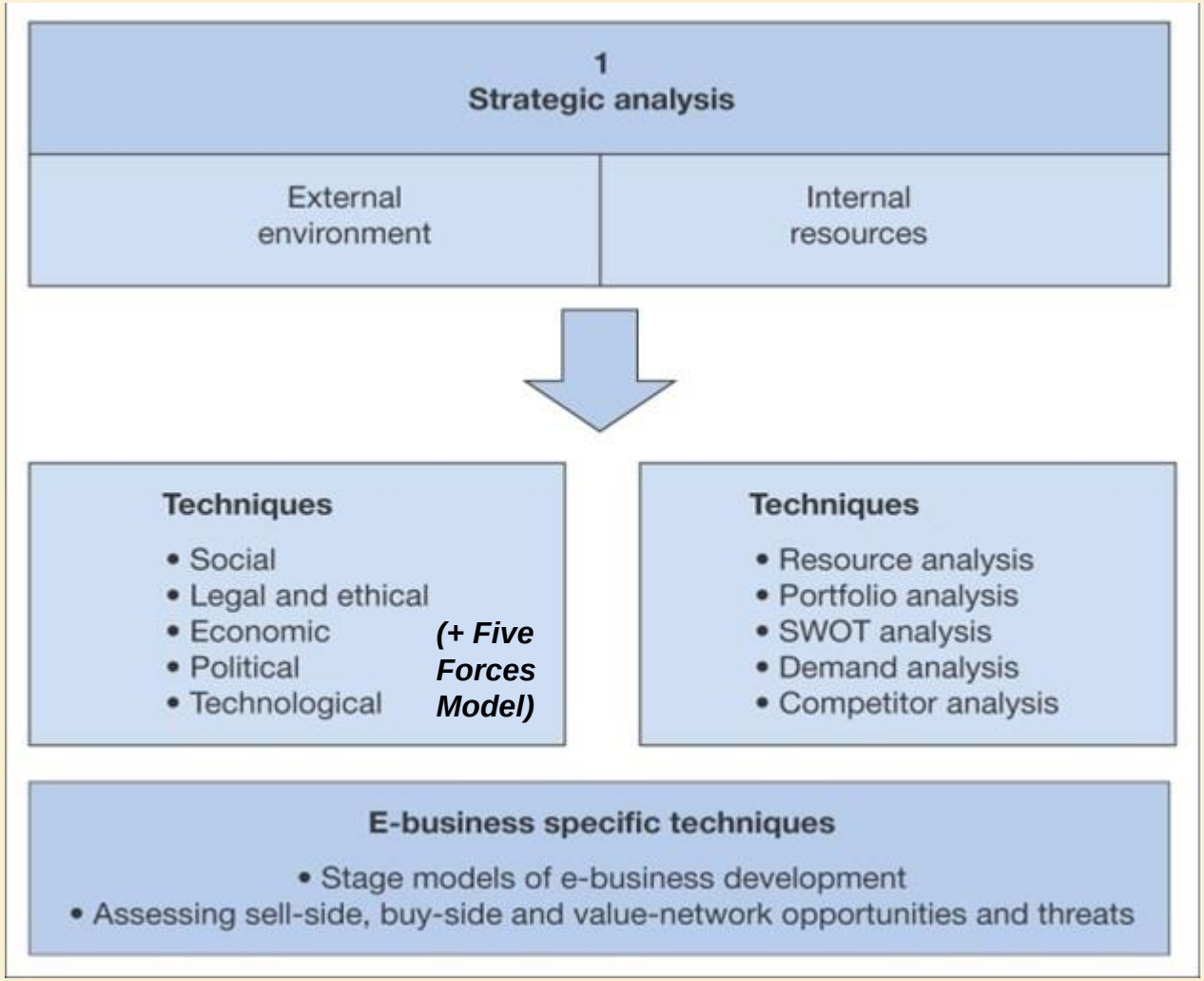


Figure 3: Elements of strategic situation analysis for the e-business.

- Collection and review of information about the external environment and internal resources
  - Immediate competition
  - Wider environment
  - Internal resources

Stage models of E-business development

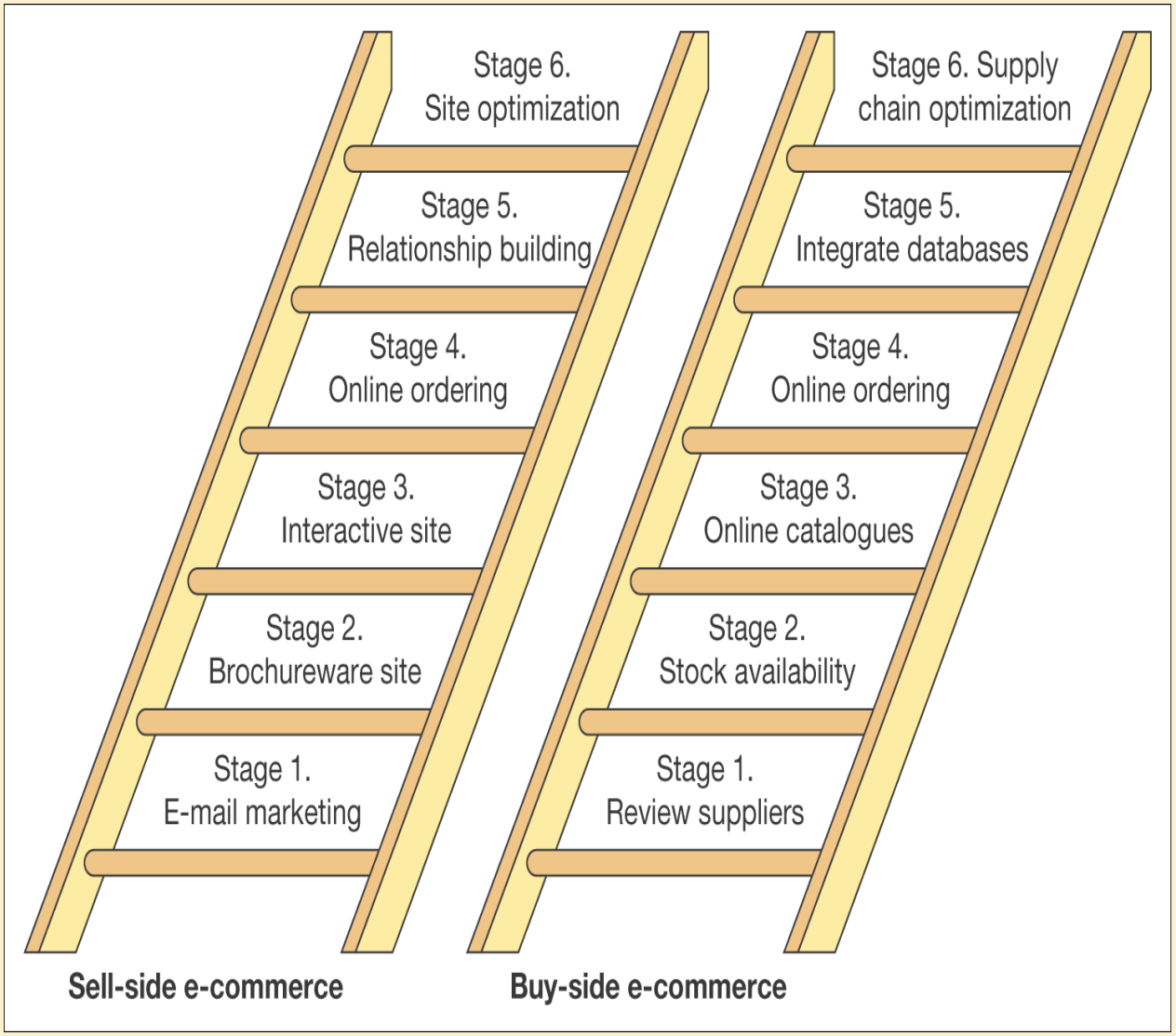


Figure 4: A simple stage model for buy-side and sell-side e-commerce.

- A stage model was presented, which could be helpful in assessing the business's position and resources

Portfolio analysis

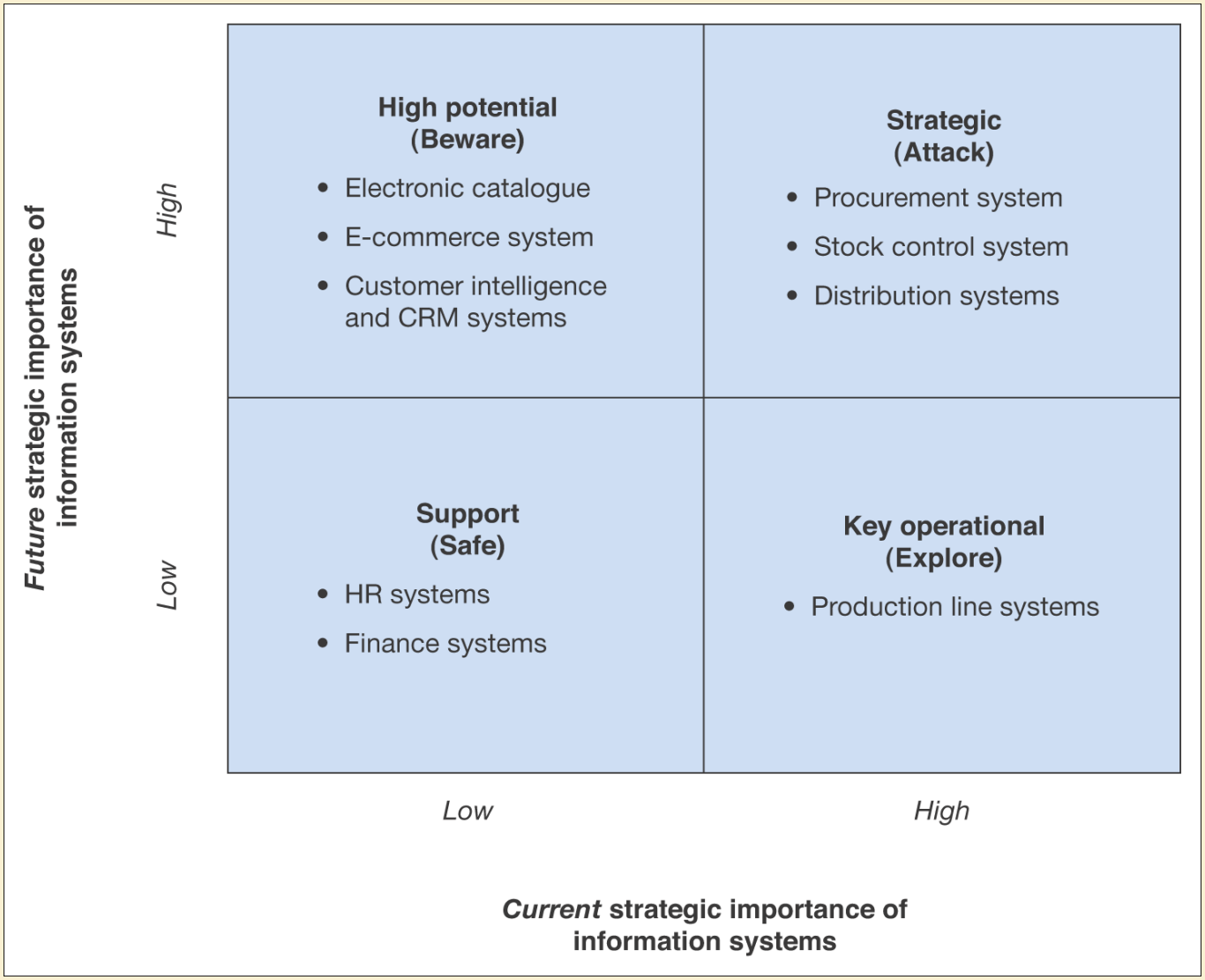


Figure 5: Summary applications of a portfolio analysis for an example B2B company.

- Portfolio analysis is helpful for assessing current e-business capability and future e-business strategies

SWOT Analysis

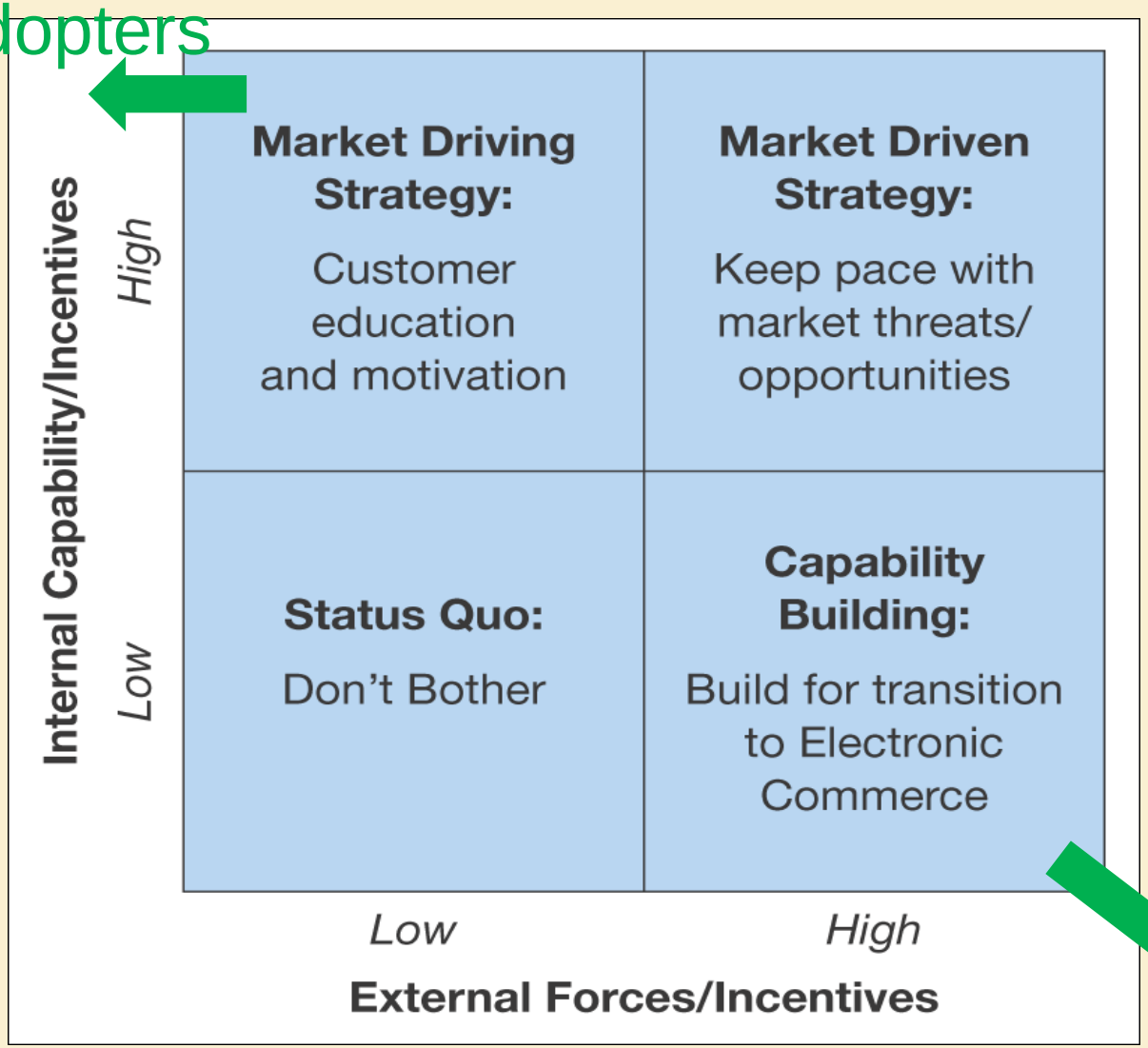
The organisation	Stengths – S 1. Existing brand 2. Existing customer base 3. Existing distribution	Weaknesses – W 1. Brand perception 2. Intermediary use 3. Technology/skills 4. Cross-channel support
Opportunities – O 1. Cross-selling 2. New markets 3. New services 4. Alliances/co-branding	SO strategies Leverage strengths to maximise opportunities = Attacking strategy	WO strategies Counter weaknesses through exploiting opportunities = Build strengths for attacking strategy
Threats – T 1. Customer choice 2. New entrants 3. New competitive products 4. Channel conflicts	ST strategies Leverage strengths to minimise threats = Defensive strategy	WT strategies Counter weaknesses and threats = Build strengths for defensive strategy

Figure 6: Example of E-business SWOT analysis.

- Strengths, weaknesses, opportunities, and Threats analysis is a useful tool for analyzing the current situation and for formulating strategies

Balance Internal and External

Early adopters



- The adoption of an E-business strategy will be based on the balance between internal capability and external forces

A later adopter

Figure 7: Matrix for evaluation of external capability against internal capability for different options based on such a balance.

Competitive Environment Analysis
Porter's five forces

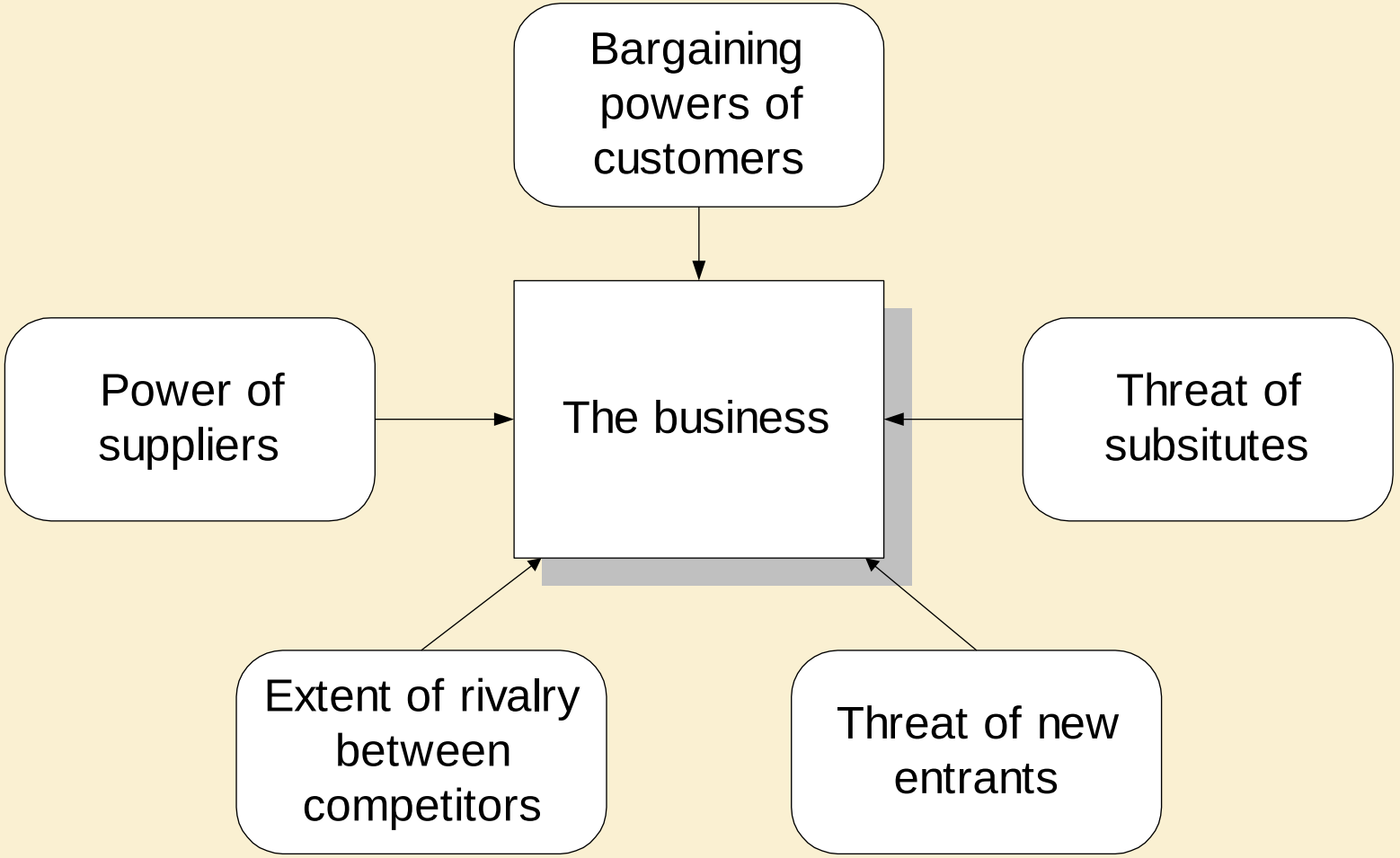


Figure 8: Competitive Environment Analysis Porter's five forces.

Elements of strategic objective setting for the E-business

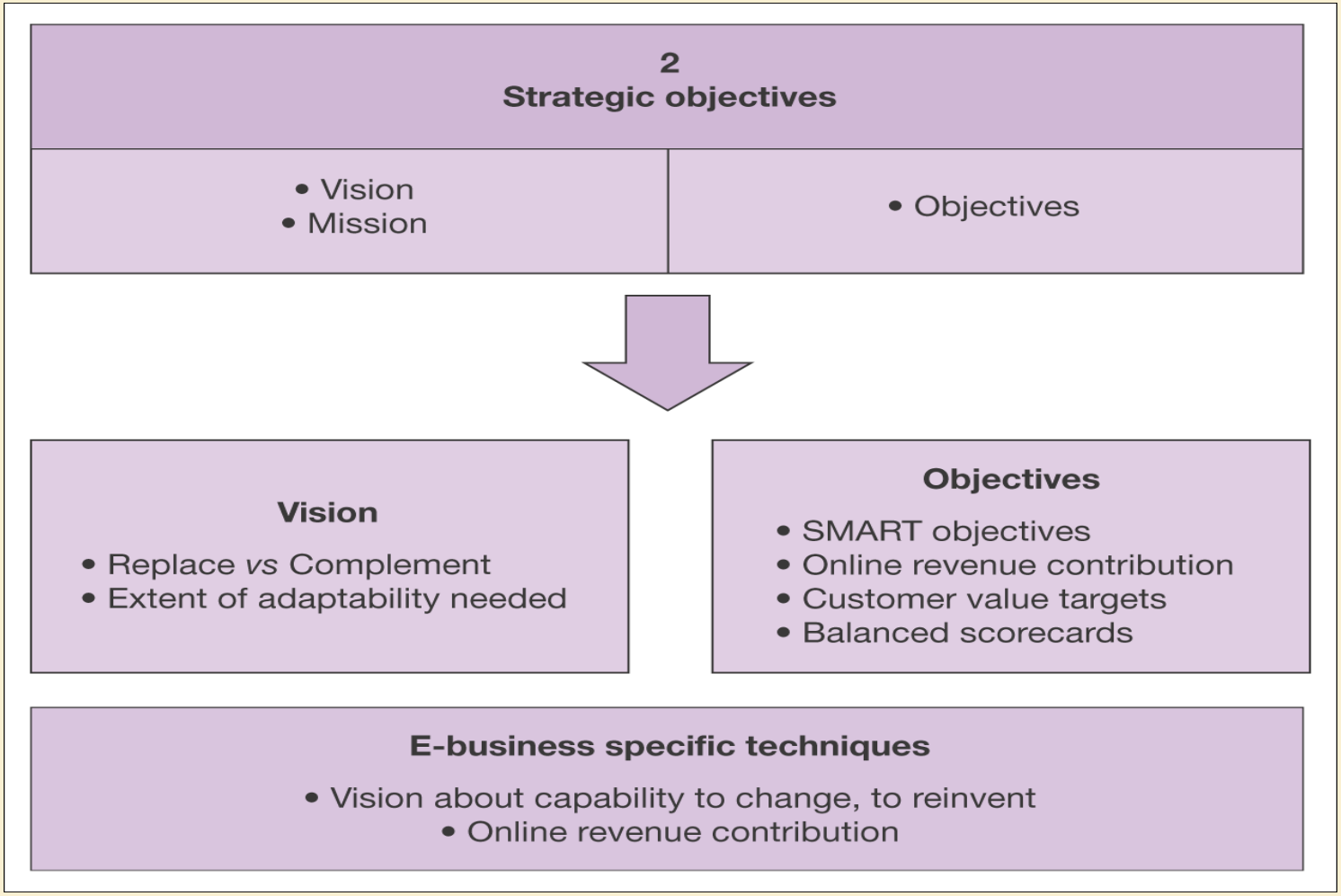


Figure 9: Elements of strategic objective setting for the E-business.

How can e-business create value

- Following methods would be proposed
  - Adding values—better quality products and services
  - Reducing cost—making the business process more efficient
  - Managing risks—improve information flow and availability
  - Creating new reality—create new way that products and or services
  - …

An evaluation tool relating information to business value

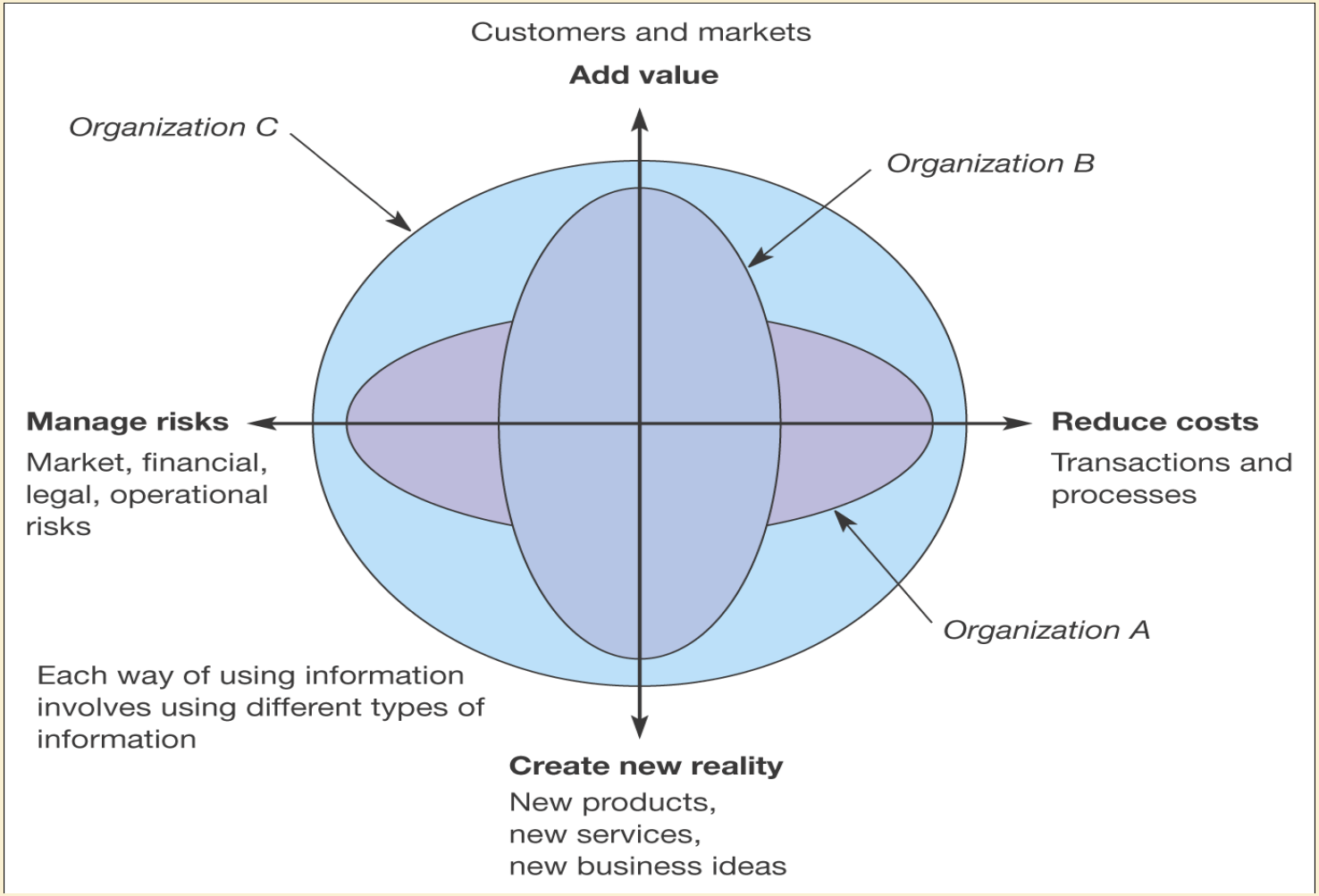


Figure 10: An evaluation tool relating information to business value.

SMART Objectives

- The 2nd phase of strategy development process will produce objectives.
- Online Revenue Contribution--% of company revenue directly generated through online transactions
- B2B should consider indirect revenue
- The objectives should be SMART
  - Specific
  - Measurable
  - Actionable/Achievable
  - Relevant
  - Time-Related

SMART Objectives

- **Specific**: Is the objective sufficiently detailed to measure real-world problems and opportunities?
- **Measurable**: Can a quantitative or qualitative attribute be applied to create a metric?
- **Actionable**: Can the information be used to improve performance? If the objective doesn't change behavior in staff to help them improve performance, there is little point in it!
- **Relevant**: Can the information be applied to the specific problem faced by the manager?
- **Time-related**: Does the measure or goal relate to a defined timeframe?

Objectives, strategies and performance indicators for an example B2B company

Objectives	Strategies of achieve goals	Key performance indicators (critical success factors)
1. Develop revenue from new geographical markets.	1. Create an e-commerce facility for standard products and assign agents to these markets.	1. Achieve combined revenue of £1m by year-end. Online revenue contribution of 70%.
2. Increase revenue from smaller-scale purchases from retailers.	2. Create e-commerce facility for standard products.	2. Increase sales through retailers from 15% to 25% or total by year 2. Online revenue contribution of 30%.
3. Improve efficiency of sourcing raw materials.	3. Develop e-procurement system.	3. Reduce cost of procurement by 5% by year-end, 10% by year 2.
4. Reduce time to market and costs for new product development.	4. Use collaboration and project management tools.	4. Reduce cost and time to market by average of 10% by year 3.
5. Protect and increase efficiency of distributor and partner network.	5. Create partner extranet and aim for paperless support.	5. Reduce cost of sales in each of five main geographical markets by 30%.

Table 1: Objectives, strategies and performance indicators for an example B2B company.

Scorecard for Objective Setting

- Scorecard is a comprehensive framework for setting and monitoring business performance.
- Metrics includes
  - Customer issues
  - Internal efficiency
  - Financial, and
  - Innovation

Example of Balanced Scorecard for Objective Setting

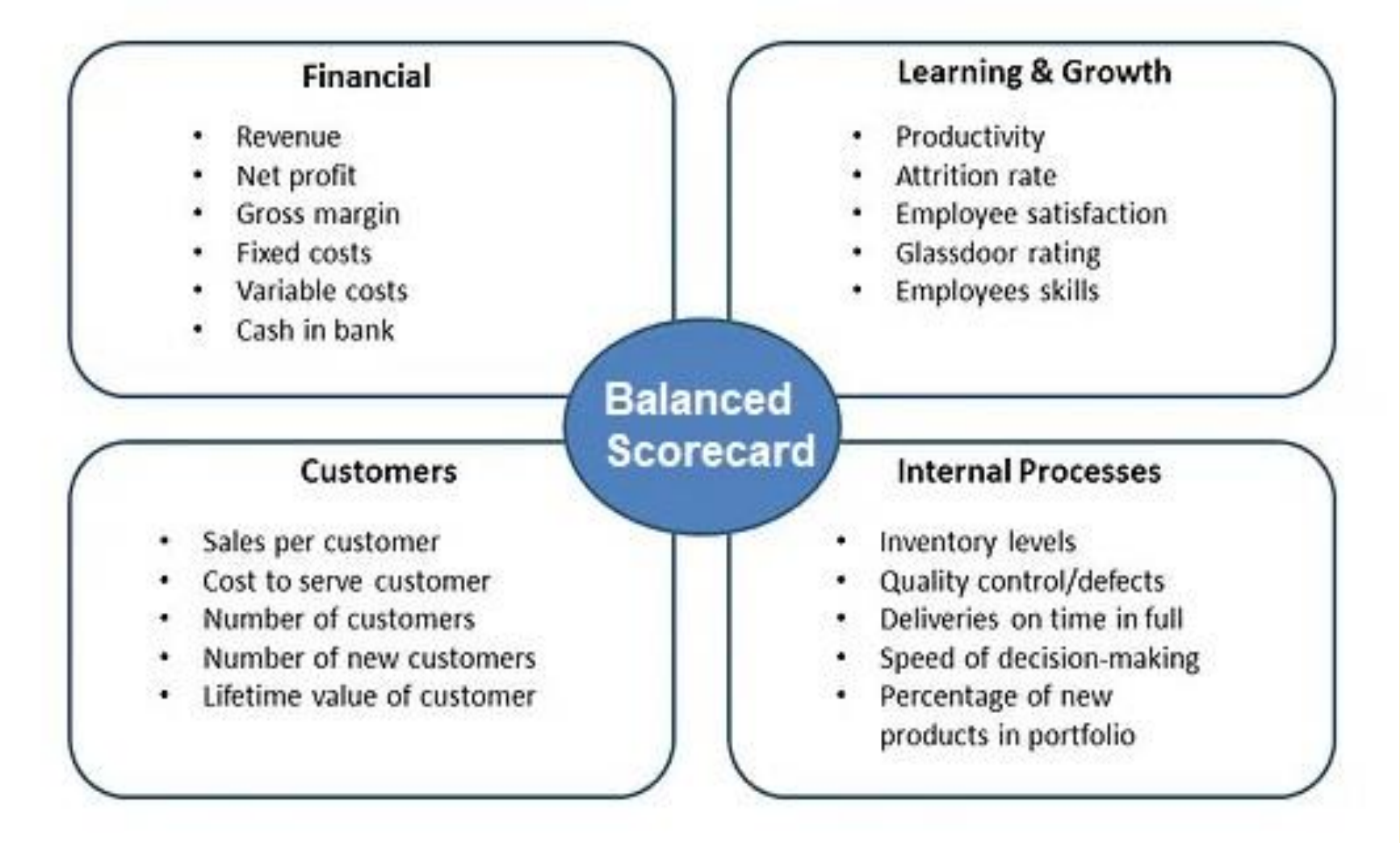


Figure 11: Example of Balanced Scorecard for Objective Setting. Source: https://www.b2bframeworks.com/balanced-scorecard.

Example of Balanced Scorecard for Objective Setting

Set objectives, measures, targets and initiatives for each quadrant - Financial, Learning/Growth, Customers, and Internal Processes. For example:

Objective	Measure	Target	Initiative
Increase revenue	Sales revenue in segment	$5.3 million	Promotion of new product
Increase profitability	Net profit	$0.6 million	Manage fixed costs
Increase profitability	Gross margin	Raise margin from 35% to 40%	Increase prices 3%

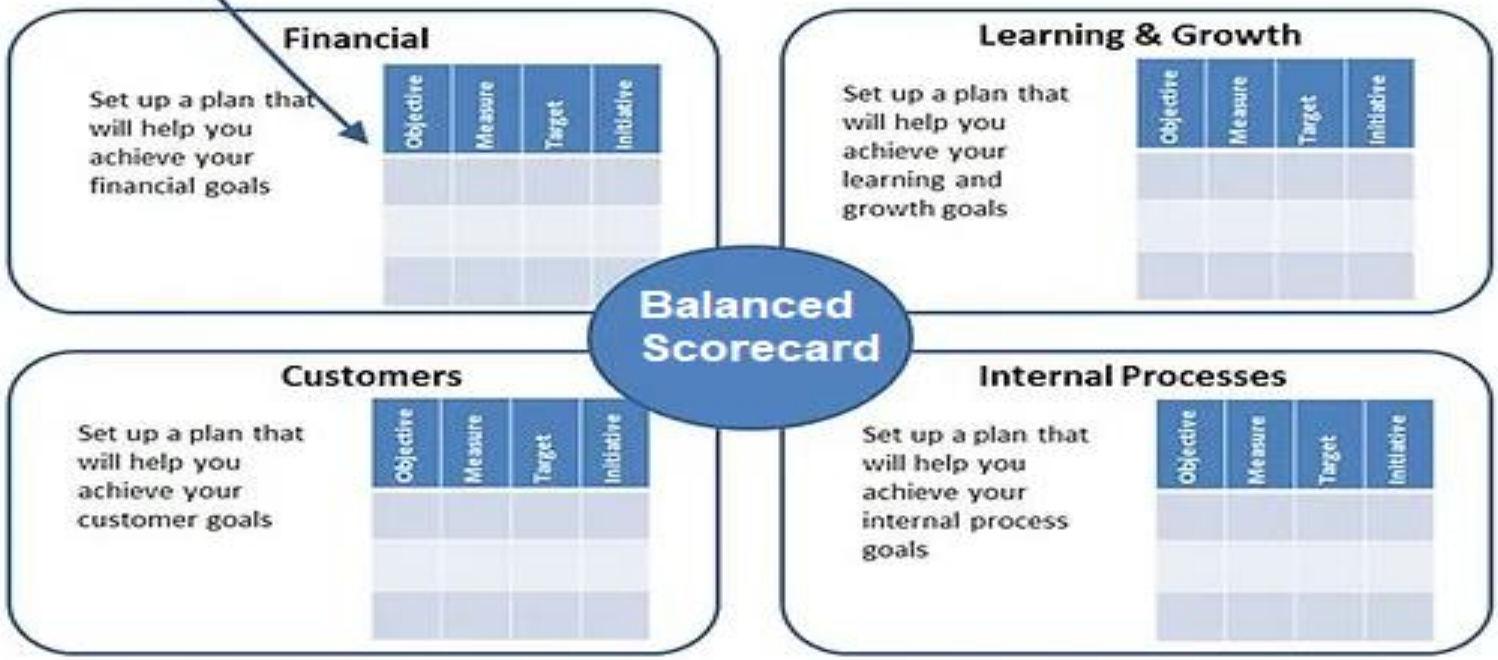


Figure 12: Example of Balanced Scorecard for Objective Setting. Source: https://www.b2bframeworks.com/balanced-scorecard.

Strategy definition

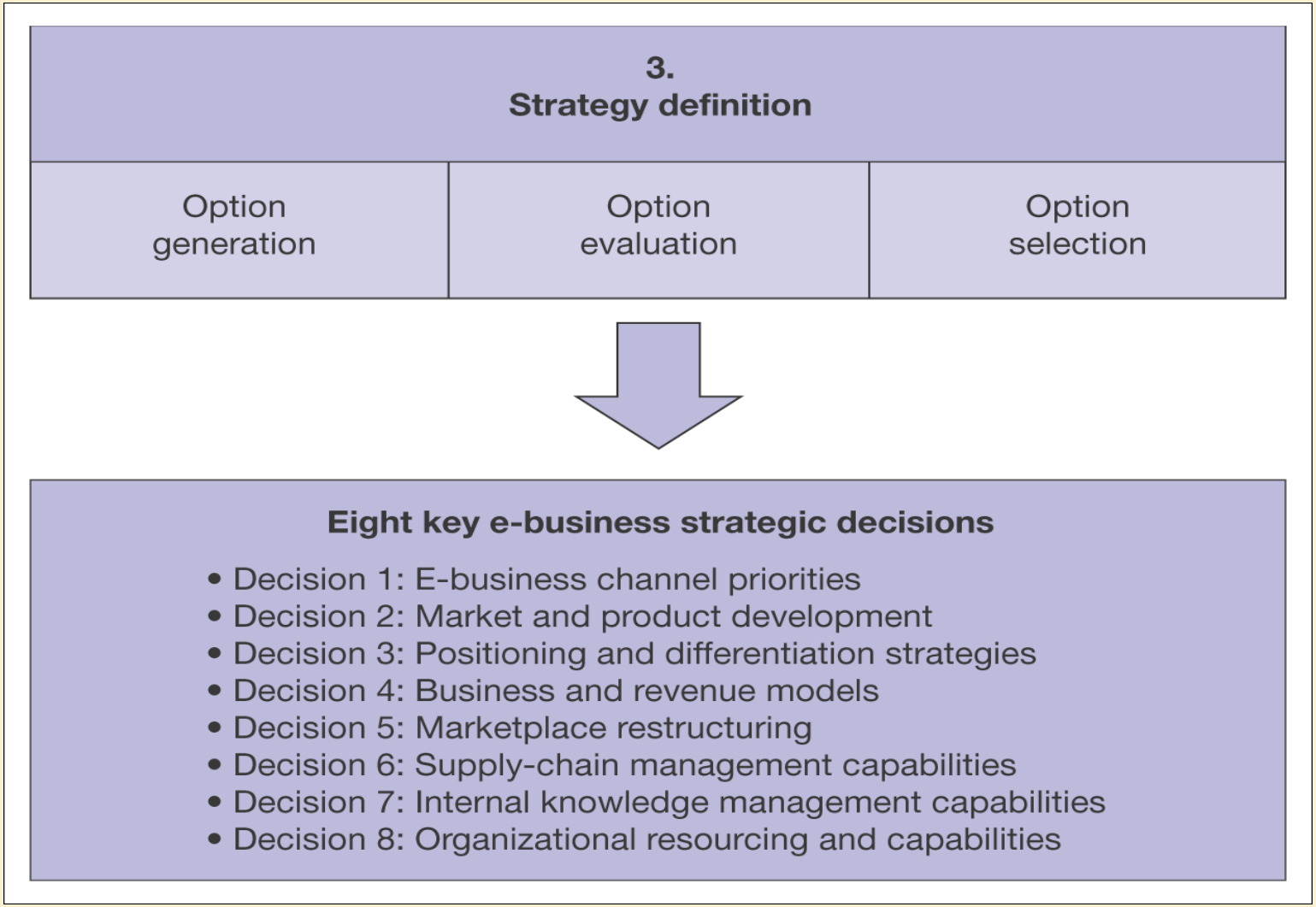


Figure 13: Elements of strategy definition for the e-business.

Decision 1: E-business channel priorities

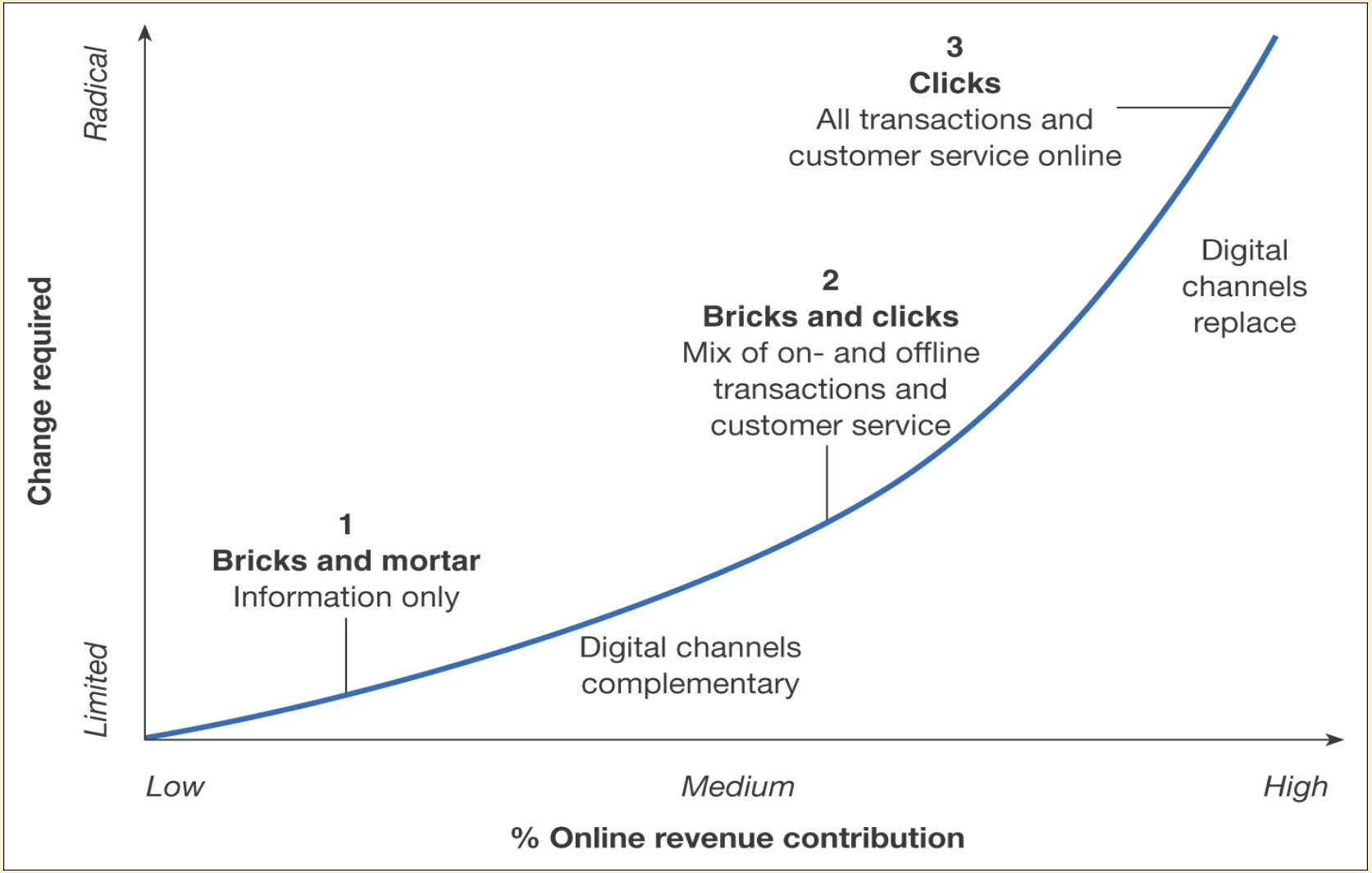


Figure 15: Strategic options for a company in relation to the importance of the Internet as a channel.

Decision 1: E-business channel priorities (Cont.)

- ***Right-channeling***
  - It involves devising a contact strategy and tactics which integrate different channels, supported by technology to reach:
    - The right person
    - At the right time
    - Using the right communications channel
    - With a relevant offer, product or message

Decision 2: Market and Product Development

- A key strategic considering is which market to target and using what products

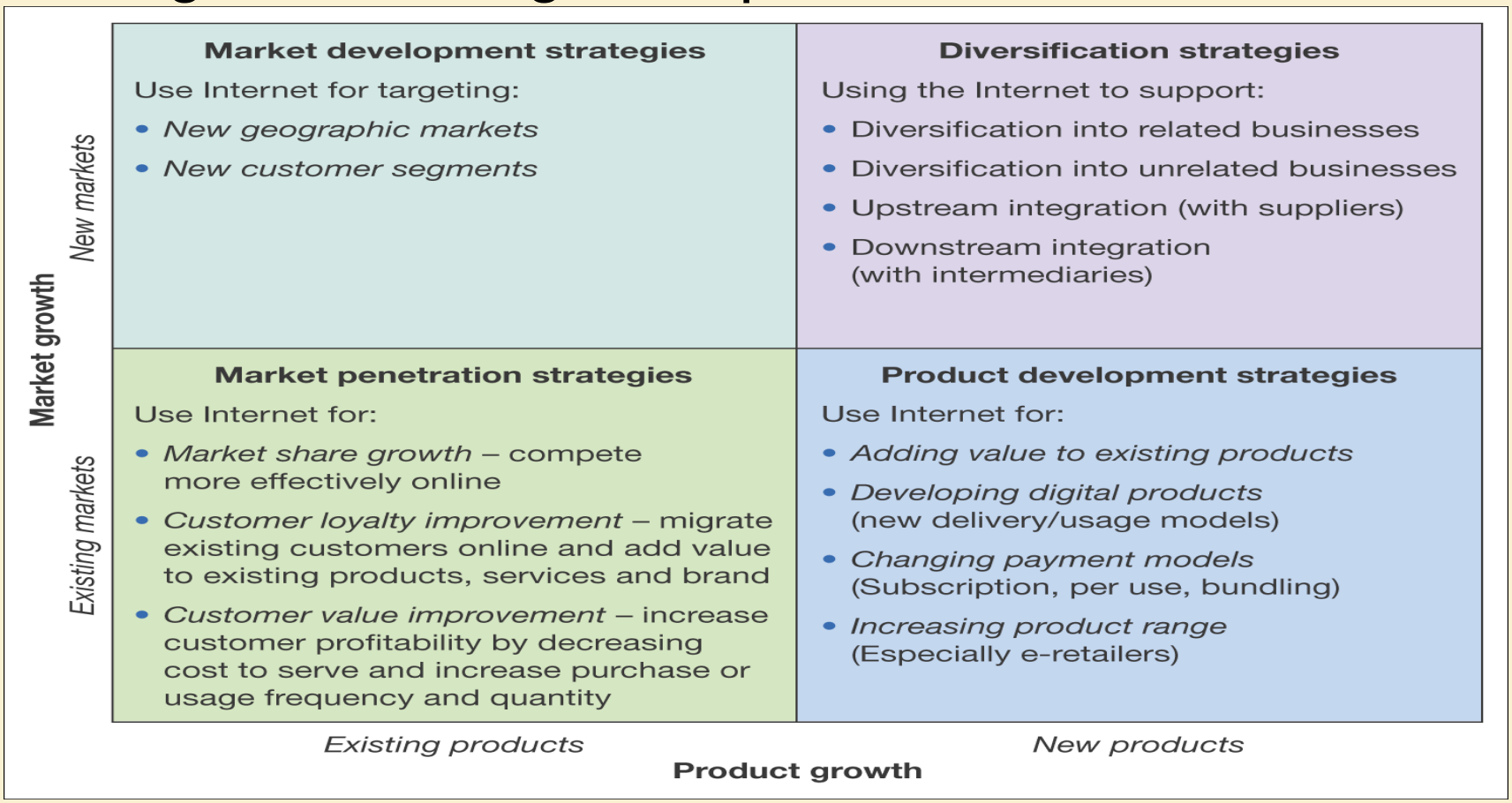


Figure 16: Using the Internet to support different growth strategies.

Decision 3: Positioning and Differentiation

- Once the market is identified, we need to define the best position of our e-commerce services relative to competitors.
- Research suggests that ***customer brand perception*** is determined by this formula
  - *Customer Value = (product quality * service quality)/ (price * fulfillment time)*
- Strategies should review the extent to which increases in product and service quality can be matched by decreases in price and time.

Decision 4:
Business, Service and Revenue Model

- ***Business model***
  - *A summary of how a company will generate revenue, identifying its product offering, value added services, revenue sources and target customers.*
- ***Revenue models***
  - *Describe methods of generating income for an organization.*

Decision 4: Business, Service and Revenue Model (Cont.)

- This decision is closely related or similar to decision 3.
- However, here it emphasizes new models, e.g., Amazon
  - *"Only the paranoid will survive" (Andy Grove)*
- Also, need to notice that many companies are successful by sticking to a single business model not far from their original vision.

Decision 5: Marketplace Restructuring

- Companies should also look at the new opportunities created by the Internet within a marketplace in this regard
  - *Disintermediation*: The removal of intermediaries such as distributors or brokers that formerly linked a company to its customers.
  - *Re-intermediation*: The creation of new intermediaries between customers and suppliers providing services such as supplier search and product evaluation.
  - *Counter-mediation*: Creation of a new intermediary by an established company.

Decision 6: Supply Chain Management Capabilities

- This might be considered together with decision 5. The main e-business strategy decisions that need to be reviewed are:
  - How should we integrate more closely with our suppliers, for example through creating an extranet to reduce costs and decrease time to market?
  - Which types of materials and interactions with suppliers should we support through e-procurement?
  - Can we participate in online marketplaces to reduce costs?

Decision 7:
Internal Knowledge Management Capabilities

- What is knowledge?
  - Patterns, rules, and contexts provide a framework for creating, evaluating, and using information.
  - It can be tacit (undocumented) or explicit (documented) knowledge
- What is knowledge management?
  - Set of processes to create, store, transfer, and apply knowledge.

Decision 8: Organizational Resources and Capabilities

- Once the e-business strategy decisions have been reviewed and selected, decisions are needed on how the organization needs to change to achieve the priorities set for e-business.
- Available options
  - In-house division (integration)
  - Joint venture (mixed)
  - Strategic partnership (mixed)
  - Spin-off (separation)

Decision 8: Organizational Resources and Capabilities (Cont.)

- Other changes to be examined
  - Strategy process and performance improvement
  - E-commerce infrastructure
  - Senior management support
  - Market integration
  - Online market focus
  - Etc.

Strategy Implementation

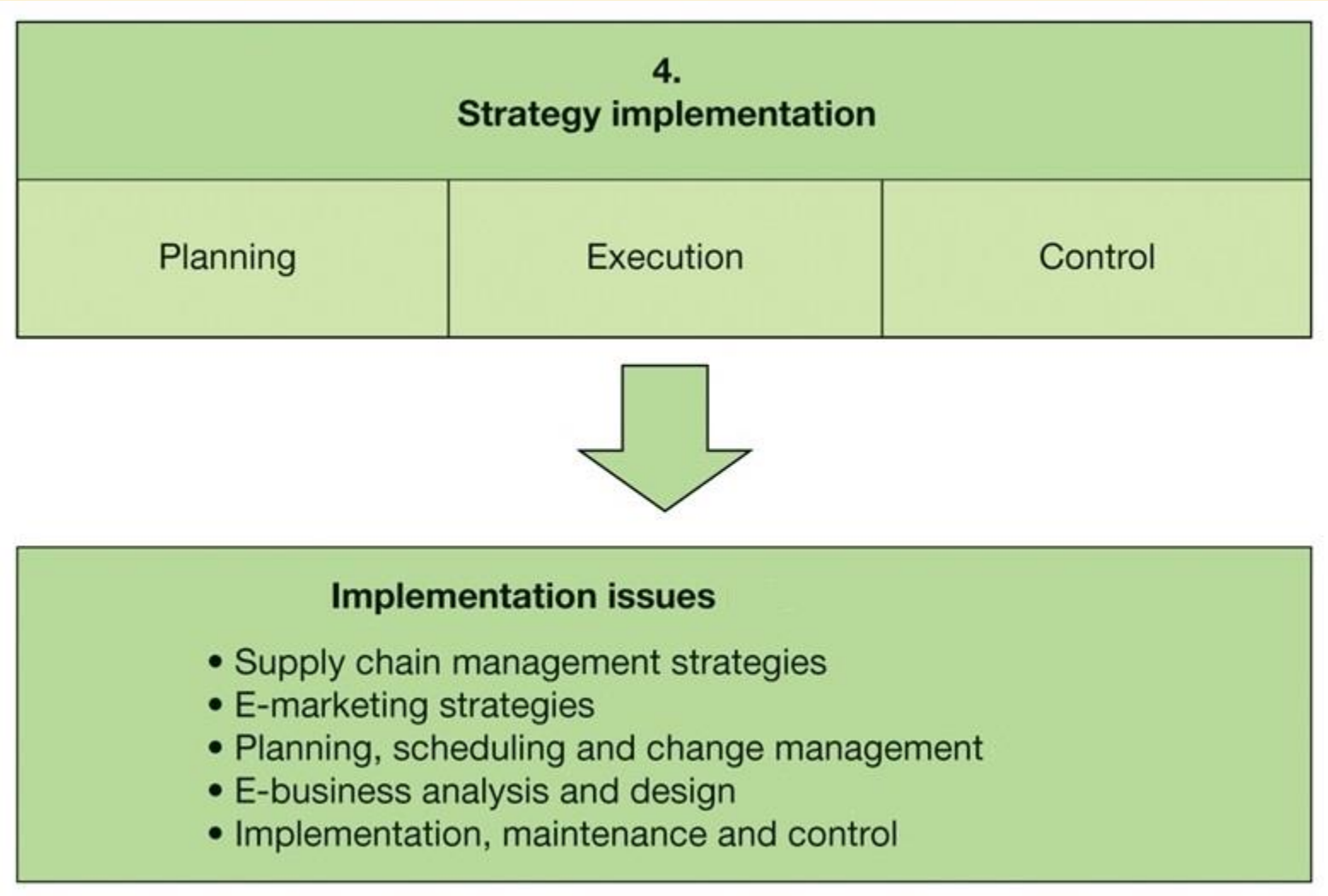


Figure 17: Elements of strategy implementation for the E-business.

Failed E-business Strategies

- Reasons
  - Overestimate the speed that the marketplace adopts the dot.com innovation
  - Timing errors
  - Lack of creativity
  - Free services
  - Over ambition
  - Etc.

Failed E-business Strategies (Cont.)

- Usually there are more fundamental problems than appeared
- Major problems occurred during the strategy development and implementation process
  - *Situation Analysis*: Insufficient research on demand
  - *Object setting*: No objectives or unrealistic
  - *Strategy definition*: Poor business models
  - *Implementation*: Problems with customer services and product quality, etc.

Implementation Success Factors for SMEs

- Effective
- Convenience
- Manageable and under control
- Brand image
- Commitment
- Partnership
- Process Improvement
- …

Information Systems Strategy and e-Business Strategy

- These two are closely related.
- Business-alignment approach—top down.
  - Start with business strategy and make IS strategy so that it aligns with business strategy.
- Business-impact approach—bottom up.
  - Examine new IS opportunities to see if they can bring a positive impact to business strategy.

The Productivity Paradox

- Research indicates that there is a weak or poor correlation between IS investment and business performance.
- It's agreed that IT investment has a strong positive relationship with sales, assets, and equity but not with net income.
- Investment in IT staff and user training does show a positive relationship with income.
- It's strongly recommended that more attention should be paid to business process change when implementing an e-business strategy.

In conclusion



- Definitions of strategy
- Different forms of organizational strategy
- Relationship between E-business strategy and other strategies
- A generic strategy process model
- Dynamic E-business strategy model
- Strategic situation analysis for the E-business
- Strategic objective setting for the E-business
- Strategy definition for the E-business
- Strategy implementation for the E-business.
- …

2. Web server and E-mail technologies

Web Server Software

- A web server program is software that runs on the website hosting Server computer.
- When the user clicks a hyperlink on a Web page, a request is sent to the Web server for the page associated with the link.
- It is the HTTP protocol that responds to the request and sends the results to the client machine.
- Choosing Web server software is not an easy task due to having many products available, with many different features.

Web Servers for Windows and Linux

1. Apache

- It is the second most popular web server software used by 31.5% of all known websites.
- It's an open-source project that uses HTTP protocol and operates across various OSs, including Windows and Linux.
- Key Features for Apache
  - IPv6
  - Session tracking
  - FTP and HTTP/2
  - Customizable modules

Source: https://www.elegantthemes.com/blog/wordpress/best-web-servers-for-windows-and-linux#8-microsoft-iis

Web Servers for Windows and Linux

1. Apache

- Apache is for you if…
  - You're a beginner.
  - You're looking for open-source and customizable server software.
  - You run a low to medium traffic WordPress site.

Web Servers for Windows and Linux (Cont.)

2. Microsoft IIS

- It is an excellent server software option that's specifically designed for Windows.
- It includes many native Windows security features, such as Azure Active Directory.
- Additionally, it has integrated website and server management tools.
- The software includes native support for dynamic ASP.NET applications, spanning CSS, JavaScript, and HTML.
- Microsoft IIS isn't compatible with Linux.

Web Servers for Windows and Linux (Cont.)

2. Microsoft IIS

- Key Features for Microsoft IIS
  - Integrations with many Microsoft products
  - Developer-friendly
  - Simple and user-friendly interface
  - Extensive built-in security features
- Microsoft IIS is for you if…
  - You use Windows.
  - You're a developer.
  - You want to run multiple sites on one server.

Web Servers for Windows and Linux (Cont.)

3. Tomcat

- It is one of the best web server software options for Java applications.
- It uses multiple Java specifications in an open-source environment that's optimized for speed and performance.
- Key Features for Tomcat
  - Customizable modules
  - Multiple Java technologies, including Jakata WebSocket
  - Performance-enhanced data processing
  - Open-source design

Web Servers for Windows and Linux (Cont.)

3. Tomcat

- Tomcat is for you if……
  - You work primarily with Java applications and dynamic content.
  - You're looking for lightweight server software.

Web Servers for Windows and Linux (Cont.)

4. NGINX

- It's recently overtaken Apache and is currently used by more than 33% of known websites.
- In addition, NGINX is compatible with both Linux and Windows.
- Key Features for NGINX
  - Compatibility with various web applications
  - Minimal resource consumption
  - Load balancing
  - Open-source design

Web Servers for Windows and Linux (Cont.)

4. NGINX

- NGINX is for you if…
  - You have a high-traffic website.
  - You're looking for performance-optimized web server software.
  - You'd like a cost-effective solution that uses minimal resources.

Web Servers for Windows and Linux (Cont.)

5. LiteSpeed

- It is lightweight and security-focused web server software.
- It's designed to provide top performance and scalability with minimal usage of your server resources.
- Plus, it has built-in protection against Distributed Denial of Service (DDoS) attacks.

Web Servers for Windows and Linux (Cont.)

5. LiteSpeed

- Key Features for LiteSpeed
  - Event-driven architecture
  - Compatibility with multiple hosting panels
  - Adjustable security features
  - PHP optimization
- LiteSpeed is for you if…
  - Speed and performance are high priorities.
  - You run a high-traffic or e-commerce website.
  - You use Linux.

Web Servers for Windows and Linux (Cont.)

6. CentOs Stream

- If you need an environment to test and develop applications, you might benefit from using CentOs Stream.
- It's a Linux distribution with a built-in development platform where you can trial your applications.
- CentOS is only usable with Linux.

Web Servers for Windows and Linux (Cont.)

6. CentOs Stream

- Key Features for CentOs Stream
  - Open source and free
  - Preview of RHEL updates
  - Developer friendly
- CentOs Stream is for you if…
  - You're a developer and work with RHEL.
  - You work exclusively with Linux.
  - You're looking for free web server software.

Web Servers for Windows and Linux (Cont.)

7. Caddy

- It is a user-friendly and secure web server. This software uses HTTPS connections by default.
- It can be an excellent choice to encrypt your website's data and protect it against malicious interceptions..
- It is compatible with both Linux and Windows OS. It also supports macOS, Solaris, and BSD.
- It is a great choice for administrating multiple websites on one server, since it has a streamlined interface.

Web Servers for Windows and Linux (Cont.)

7. Caddy

- Key Features for Caddy Stream
  - Support for static and dynamic content
  - Load balancing features
  - Focus on security
- Caddy is for you if…
  - You're looking for a user-friendly solution.
  - Security is a top priority.
  - You run multiple sites.

Web Servers for Windows and Linux (Cont.)

8. Lighttpd

- It is one of the best web server software options for low-resource environments.
- It uses minimal memory and includes speed optimizations to take pressure off of the physical server.
- Furthermore, it is designed to run on Linux. You can also use it on Windows, although it lacks native support.
- In addition, It offers support for various programming languages, including PHP, Python, Lua, and Ruby..

Web Servers for Windows and Linux (Cont.)

8. Lighttpd

- Key Features for Lighttpd
  - Load balancing and HTTP proxy
  - Focus on efficiency and speed
  - Lightweight design
  - HTTP/2
- Lighttpd is for you if…
  - You consider speed and resource consumption.
  - You work with multiple programming languages.
  - You're looking for secure and flexible server software.

Web Server Hardware

- Web server hardware is a computer that is used to host a website.
- All html files, databases and image files that make up the entire content of the website are stored on the server.
- Web servers run on a Windows or Linux or other operating systems and use web server software to manage access requests to the website.
- Web server hardware is one of the major components of the E-commerce infrastructure on which the performance of the whole E-commerce application depends.

Web Server Hardware (Cont.)

- Managing and maintaining a web server is very important.
- Therefore, most of the small and medium sized business companies will pay for hosting instead of maintaining their own server.
- Factors determining web server hardware and software requirements.
  - Size of the web site
  - Purpose of the web site
  - Traffic on the web site

Web Server Hardware (Cont.)

- Web Server Hardware Requirements of B2B and B2C commerce sites include:
  - They must be available 24 hours a day, 7 days a week
  - Reliable servers
  - Backup servers for high availability
  - Efficient and easily upgraded software
  - Security software
  - Database connectivity

Web Server Hardware (Cont.)

- The following parts of the computer make bigger impact on the performance of the e-Commerce
  - Network card
  - Server Processor
  - Server Memory
  - Hard Drives
  - …

Basic Functionality Provided by Web Servers

Functionality	Description
Processing of HTTP requests	Receive and respond to client requests for HTML pages
Security services (Secure Sockets Layer)/ Transport Layer Security	Verify username and password; process certificates and private/public key information required for credit card processing and other secure information
File Transfer Protocol	Permits transfer of very large files from server to server

Basic Functionality Provided by Web Servers (Cont.)

Functionality	Description
Search engine	Indexing of site content; keyword search capability
Data capture	Log file of all visits, time, duration, and referral source
E-mail	Ability to send, receive, and store e-mail messages
Site management tools	Calculate and display key site statistics, such as unique visitors, page requests, and origin of requests; check links on pages

E-mail Server

- *Electronic mail* (E-mail) is a computer-based application for the exchange of messages between users.
- Email server is essentially a computer system that sends and receives emails.
- A worldwide e-mail network allows people to exchange e-mail messages very quickly.
- E-mail is the electronic equivalent of a letter but with advantages in timeliness and flexibility.
- Every email that is sent has to pass through a series of mail servers on the way to its intended recipient.

Source: https://jatheon.com/blog/email-server-software/

E-mail Server Software

- Standard E-mail protocols (SMTP, IMAP, & POP3).
  - Simple Mail Transfer Protocol (SMTP) is an application used to send, receive, and relay outgoing emails between senders and receivers.
  - Internet Message Access Protocol (IMAP) allows easy access to your email wherever you are from any device.
  - Post Office Protocol (POP3) works by contacting your email service and downloading all of your new messages from it.

Should You Host Your Own Email Server?

- Self-hosting or outsourcing hosting to a third-party provider.
  - Third-party hosting (hosting providers)
    - They are responsibility for managing and maintaining their servers and affordable.
    - Additional features of paid hosting services include spam filtering, online storage, email backup, etc.
  - Self-hosting (operating own internal mail server)
    - To avoid many of the issues found in third-party hosting such as privacy, security, and control over the email servers.

E-mail Client

- Sent emails can be accessed using a mail server in two ways such as *cloud-based email service* or *an on-premises email client*.
  - *Email service* – also known as webmail – is a platform via which we send and receive emails using a web-based interface and a web browse, such as Gmail, Yahoo, etc.
  - *Email client* – is a software that we install on our computer and we send or receive emails through an interface provided by that client's software such as Outlook, etc.

In conclusion

- Web Server Software
- Web Servers for Windows and Linux
- Web Server Hardware
- Basic Functionality Provided by Web Servers
- E-mail Server Software
- Should You Host Your Own Email Server?
- E-mail Client



THANK YOU
FOR YOUR ATTENTION

Q&A